1. Supervised Learning

Question: Design pseudocode to build a supervised learning workflow that accepts a labelled dataset, separates features and target values, trains a model, and predicts the target for new data.

Pseudocode

```
BEGIN
INPUT labelled_dataset
SEPARATE features,target
TRAIN model
INPUT new_data
PREDICT target
PRINT prediction
END
```

2. kNN Classifier

Question: Design pseudocode for a k-Nearest Neighbors (kNN) classifier that calculates the distance between a test sample and all training samples, selects the k nearest samples, and predicts the class using majority voting.

Pseudocode

```
BEGIN
INPUT training_data,test_sample,k
CALCULATE distance to all samples
SELECT k nearest
PREDICT majority class
END
```

3. kNN Classifier

Question: Design pseudocode to classify a new student as Pass or Fail using kNN based on attendance percentage, internal marks, and assignment scores.

Pseudocode

```
BEGIN
INPUT attendance,marks,assignment
CALCULATE distances
SELECT k neighbors
IF majority=Pass THEN PRINT Pass ELSE PRINT Fail
END
```

4. kNN Classifier

Question: Design pseudocode for selecting the appropriate value of k in a kNN classifier by evaluating different k values on validation data.

Pseudocode

```
BEGIN
FOR k=1 TO MaxK
 TRAIN & VALIDATE
 STORE accuracy
ENDFOR
SELECT k with highest accuracy
END
```

5. Decision Tree

Question: Design pseudocode to construct a Decision Tree classifier by selecting the best feature for splitting the training dataset and recursively creating child nodes.

Pseudocode

```
BEGIN
IF all records same class RETURN leaf
SELECT best feature
SPLIT dataset
RECURSIVELY create child nodes
END
```

6. CART

Question: Design pseudocode for the CART classification algorithm to select the best binary split using Gini impurity and construct a classification tree.

Pseudocode

```
BEGIN
CALCULATE Gini for all splits
SELECT lowest Gini split
CREATE left/right child
REPEAT until stopping
END
```

7. Decision Tree

Question: Design pseudocode to predict whether a customer will purchase a product using a Decision Tree based on age, income, previous purchases, and browsing history.

Pseudocode

```
BEGIN
INPUT age,income,purchases,browsing
APPLY decision rules
PRINT Buy or Not Buy
END
```

8. Regression Analysis

Question: Design pseudocode for a regression analysis workflow that loads a dataset, identifies independent and dependent variables, trains a regression model, and evaluates its predictions.

Pseudocode

```
BEGIN
LOAD dataset
IDENTIFY X,Y
TRAIN regression model
PREDICT values
EVALUATE model
END
```

9. Linear Regression

Question: Design pseudocode to implement simple linear regression for predicting house price from house area using the least-squares approach.

Pseudocode

```
BEGIN
INPUT area,price
CALCULATE slope & intercept
PREDICT house price
END
```

10. Linear Regression

Question: Design pseudocode to implement multiple linear regression for predicting a student's final mark using attendance, internal marks, assignment marks, and study hours.

Pseudocode

```
BEGIN
INPUT attendance,internal,assignment,hours
TRAIN MLR model
PREDICT final mark
```

END

11. Linear Regression

Question: Design pseudocode to train a linear regression model, calculate predicted values and residuals, and determine the Mean Squared Error (MSE) and R^2 score.

Pseudocode

```
BEGIN
PREDICT values
CALCULATE residuals
COMPUTE MSE
COMPUTE R²
PRINT results
END
```

12. Logistic Regression

Question: Design pseudocode for logistic regression to predict whether a student will pass or fail based on study hours and attendance percentage.

Pseudocode

```
BEGIN
INPUT study_hours,attendance
TRAIN logistic model
PREDICT Pass/Fail
END
```

13. Logistic Regression

Question: Design pseudocode that calculates the sigmoid probability for logistic regression and converts the probability into a binary class using a threshold of 0.5.

Pseudocode

```
BEGIN
CALCULATE z
prob=1/(1+e^-z)
IF prob>=0.5 THEN Class=1 ELSE Class=0
END
```

14. Logistic Regression

Question: Design pseudocode to evaluate a logistic regression classifier using the confusion matrix, accuracy, precision, recall, and F1-score.

Pseudocode

```
BEGIN
CALCULATE TP,TN,FP,FN
COMPUTE Accuracy,Precision,Recall,F1
PRINT metrics
END
```

15. Unsupervised Learning

Question: Design pseudocode for an unsupervised learning process that accepts an unlabelled dataset, preprocesses the features, discovers groups, and produces cluster assignments.

Pseudocode

```
BEGIN
LOAD unlabelled data
PREPROCESS features
DISCOVER clusters
OUTPUT assignments
END
```

16. K-Means Clustering

Question: Design pseudocode to implement the K-Means clustering algorithm: initialize centroids, assign observations to the nearest centroid, update centroids, and repeat until convergence.

Pseudocode

```
BEGIN
INITIALIZE centroids
ASSIGN nearest centroid
UPDATE centroids
REPEAT until convergence
END
```

17. K-Means Clustering

Question: Design pseudocode to group customers into three clusters based on annual income and spending score using K Means clustering.

Pseudocode

```
BEGIN
INPUT income,spending
SET K=3
RUN K-Means
PRINT cluster IDs
END
```

18. K-Means Clustering

Question: Design pseudocode to determine the suitable number of clusters using the Elbow Method and then apply K-Means with the selected number of clusters.

Pseudocode

```
BEGIN
FOR K=1 TO MaxK
 RUN K-Means
 STORE WCSS
ENDFOR
SELECT elbow K
END
```

19. Evaluation Metrics

Question: Design pseudocode to calculate TP, TN, FP, and FN from actual and predicted binary class labels and subsequently calculate accuracy, precision, recall, and F1-score.

Pseudocode

```
BEGIN
INPUT actual,predicted
CALCULATE TP,TN,FP,FN
CALCULATE Accuracy,Precision,Recall,F1
END
```

20. Model Evaluation

Question: Design pseudocode for comparing kNN, Decision Tree, Logistic Regression, and K-Means models using appropriate evaluation measures and selecting the most suitable model for a given Data Science problem.

Pseudocode

```
BEGIN
EVALUATE kNN,DT,LR,KMeans
COMPARE metrics
SELECT best model
PRINT result
```

END